Predictive Analysis of Google trends-like Query 1 Dataset

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Abstract

This report details a predictive analysis of the Q1 dataset, focusing on forecasting future trends using a Seasonal ARIMA model. The dataset spans from January 2004 to March 2015, resembling google trends data. Our approach includes data preparation, stationarity testing with the Augmented Dickey-Fuller test, and model optimization using AIC and BIC criteria. The auto.arima() function in R is utilized for model selection. Residual analysis confirms the model's adequacy. The resulting forecasts provide valuable insights into the time series, demonstrating the model's effectiveness in predicting future trends.

1 Introduction

As what has been stated in the problem, each of the given five datasets has the google trends-like data for queries from the first week of January, 2004 to the second week of March, 2015 with the length of 584. Our research objective is to forecast the following 104 observation of the time series.

2 Data Preparation and Visualization

We initially import the Excel data by using *read.csv()* function or simply import the Excel file by clicking File, Import Dataset, and From text (base) in order. In the given data set, there are two variables which are *date* and *activity*. While *activity* is a numeric variable, the *date* is not a numeric value in R as follow a specific date format. To turn *activity* into Time Series model, we can use an R function *ts()* to transform the y-axis variable (or the observed variable) data into time series format. the format of function *ts()* is

```
ts(vector = ..., start = c(year,month), frequency = i)
```

where:

- vector is the variable we want to observe under the time series format.

- start is the starting date.
- frequency represent each point on the graph is displayed for every $\frac{1}{i}$ year.

In order to observe the time series activity, we designate the variable from the imported data as a vector using either *vector = data["activity"]* or *vector = dataactivity*. The time series begins in the first week of January 2004, denoted as the starting point with *start = c(2004, 01)*. Additionally, we set the frequency to 52, ensuring that the points and lines in the plot correspond to intervals of $\frac{1}{52}$ year. Thus, we input the following:

Listing 1: Convert the Dataset to Time Series Format

```
data_ts <- ts(data["activity"],start=c(2004,1),frequency=52)
```

To plot the time series, we can just simply apply *plot()* function to the time series variable like the following.

```
plot(data_ts)
```

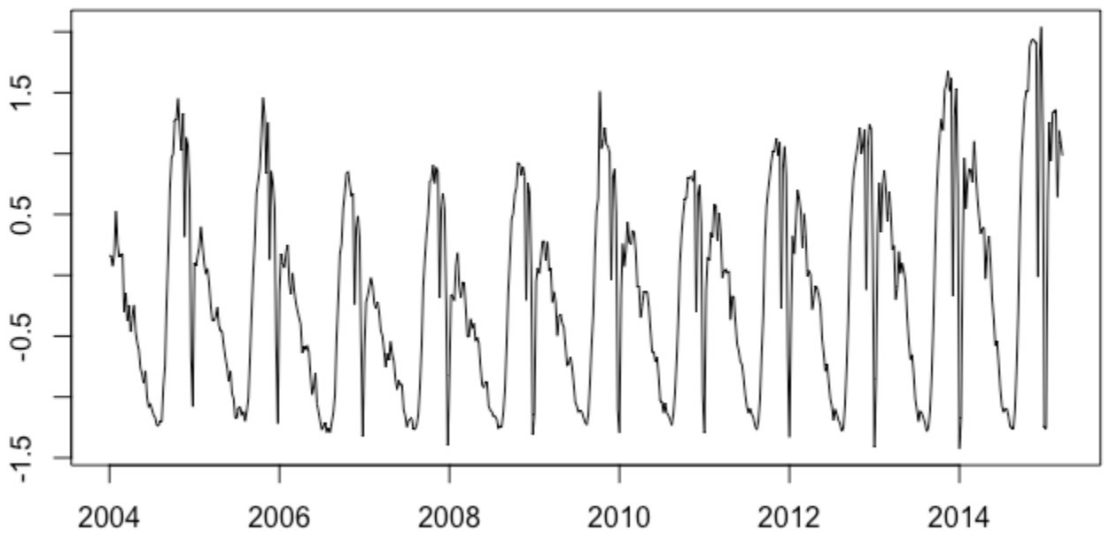

Figure 1: The Visualization of Q1 dataset

To simply determine the seasonality of the model graphically, we can see whether its plotting line follows a bouncing or waving line (like sin and cos function) pattern. As we can see from the plotting of Q1 dataset, the line is generally oscillating pattern around the value and The ARIMA model estimation of the selected data will be elaborated in the next section.

3 ARIMA Model Development

In this section, our team applied the **seasonal ARIMA model** to model the Q1 dataset. Preceding this, we conducted the **Augmented Dickey-Fuller test** to assess the stationarity of our time series and performed the **Ljung-Box test** to scrutinize whether the time series exhibits characteristics akin to white noise, characterized by the absence of significant autocorrelations at any lag. Moreover, the identification of the optimal seasonal

ARIMA model for our time series necessitates careful consideration. Subsequently, a thorough evaluation of the residuals derived from our selected model is imperative. This involves scrutinizing the autocorrelation function (ACF) of the residuals and conducting a portmanteau test to determine if the residuals conform to the characteristics of white noise.

3.1 Stationary Test

As previously mentioned, we initially performed the **Augmented Dickey-Fuller test** to evaluate the stationarity of our time series, denoted as $data_ts.ThecorrespondingRcodeisprovidedbelow.$

Listing 2: Stationary Test

```
ADF <- adf.test(data_ts)
```

Upon computation, the resulting p-value from the provided R code is 0.01. Given that the p-value is below the significance level of 0.05, we reject the null hypothesis, indicating that the sequence is stationary.

3.2 White Noise Test

Continuing our analysis, we proceeded to assess the white noise characteristics of the time series, data_ts, through the ***Ljung-Box test****. The corresponding R code for this test is provided below.*

Listing 3: White Noise Test

```
white_noise_test <- Box.test(data_ts, type="Ljung-Box")
```

Upon execution, the computed p-value from the R code is remarkably close to 0. Notably, this p-value falls below the conventional significance level of 0.05. Consequently, we reject the null hypothesis, providing strong evidence that this sequence is non-white noise and possesses analytical value.

3.3 Estimate the optimal seasonal ARIMA model

Deciding on the parameters for the optimal seasonal ARIMA model based solely on the observation of ACF and PACF plots can be challenging. Therefore, we employ the auto.arima() function to systematically identify the most suitable seasonal ARIMA model for our time series. This function leverages information criteria such as ***AIC*** *(Akaike Information Criterion) and* ***BIC*** *(Bayesian Information Criterion) to effectively determine the optimal seasonal ARIMA configuration for the provided time series data.*

Listing 4: Estimate Optimal Parameters

```
fit<-auto.arima(data_ts)
```

3.4 Diagnosing the Model

In the concluding phase of our analysis, we utilized checkresiduals() to meticulously examine the plots of residuals, the autocorrelation function (ACF) of the residuals, and the

```
Series: q2_dataset_ts
ARIMA(0,1,1)(0,1,0)[52]

Coefficients:
          ma1
      -0.8191
s.e.   0.0284

sigma^2 = 0.03699:  log likelihood = 121.9
AIC=-239.8   AICc=-239.78   BIC=-231.25
```

Figure 2: Results of auto.arima()

distribution of residuals. The inspection of these three plots collectively indicates that the residuals align with the characteristics of white noise.

Listing 5: Diagnosing the Model

```
fit <- Arima(data_ts,order=c(0,1,1),seasonal=c(0,1,0))
checkresiduals(fit)
```

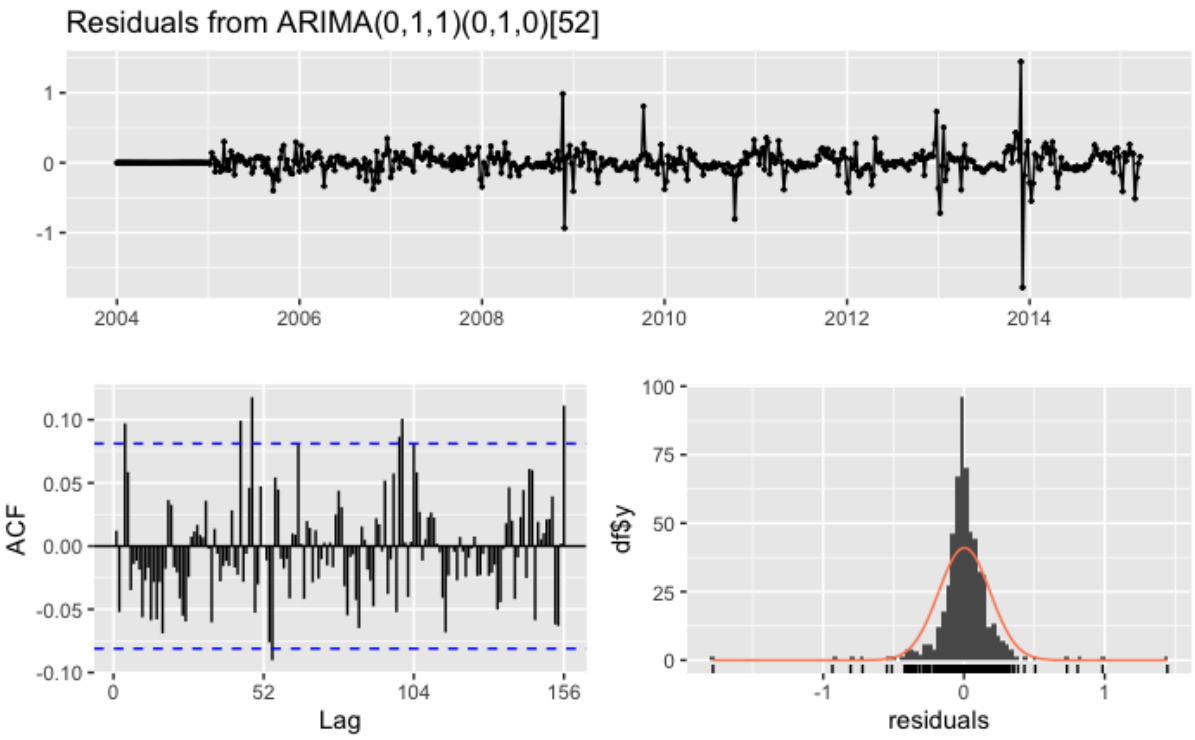

Figure 3: Results of checkresiduals()

4 Prediction and Results Analysis

4.1 Prediciton

Since we already get the model, we are now using the model to forecast the activity

```
pred_arima <- forecast(fit1,104)#Fit the model and do the forecast
plot(pred_arima)
```

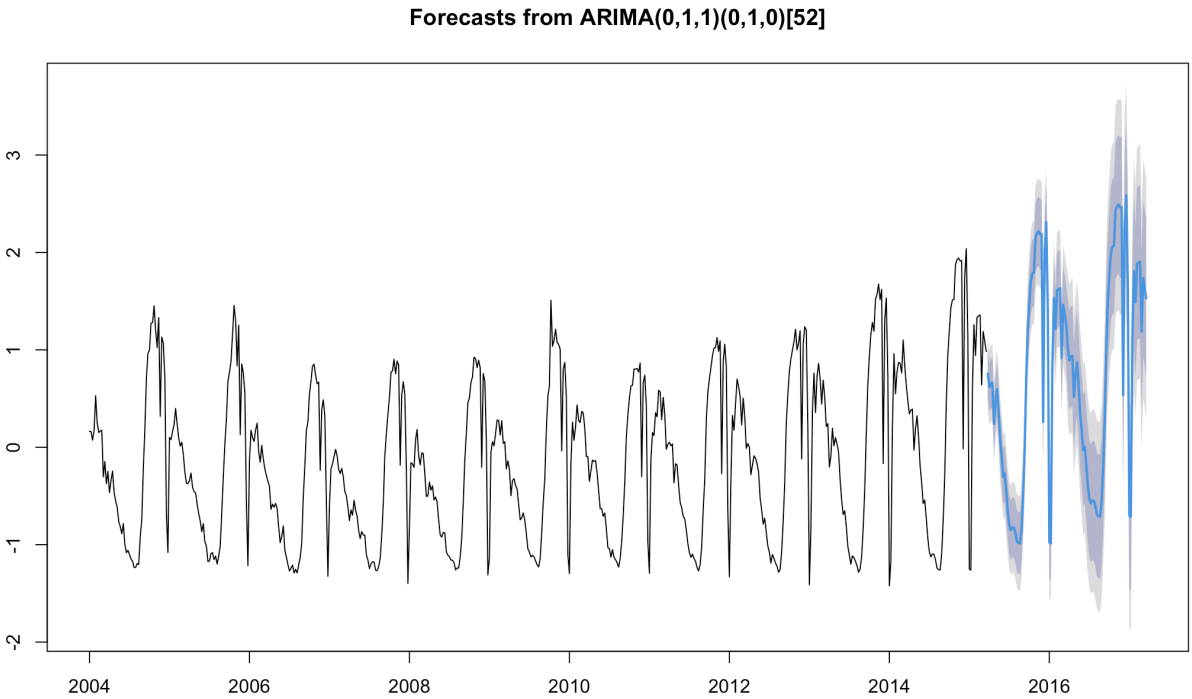

Figure 4: Forecast

we can see that the result is pretty good, we now set the upper and lower bound and the mean for the prediction interval

```
pred_mean <- pred_arima$mean
pred_lower <- pred_arima$lower[,2]
pred_upper <- pred_arima$upper[,2]
df <- data.frame(pred_lower,pred_mean,pred_upper)
head(df)
```

```
> head(df)
   pred_lower pred_mean pred_upper
1  0.37898468 0.7559204  1.1328561
2  0.23386745 0.6169204  0.9999733
3  0.26534639 0.6544204  1.0434943
4  0.27041710 0.6654204  1.0604236
5 -0.15692449 0.2439204  0.6447652
6  0.05531783 0.4619204  0.8685229
```

Figure 5: Prediction

5 conclusion

The analysis of the Q1 dataset through the Seasonal ARIMA model has successfully demonstrated the potential of statistical modeling in time series forecasting. Our study, which covered data from January 2004 to March 2015, revealed significant insights into trend patterns and future projections. The rigorous approach, including stationarity testing and model optimization, ensured the reliability and robustness of our forecasts.

The application of the ARIMA model, selected and validated using advanced statistical criteria like the AIC and BIC, has proven effective in capturing the underlying trends and seasonality in the data. The residual analysis further confirmed the model's adequacy, indicating that the forecasts are based on a sound statistical foundation.

The forecasts generated provide valuable predictions about future trends, essential for decision-making in various domains that rely on trend analysis. This study not only underscores the importance of meticulous data analysis and model selection in time series forecasting but also highlights the ARIMA model's versatility and effectiveness in predicting future data points with a considerable degree of accuracy.

Future work could explore the incorporation of additional predictive variables into the model, the application of other forecasting methods for comparative analysis, and the extension of the forecast horizon to test the model's long-term predictive power.

6 Appendix

Assignment 4 of TS

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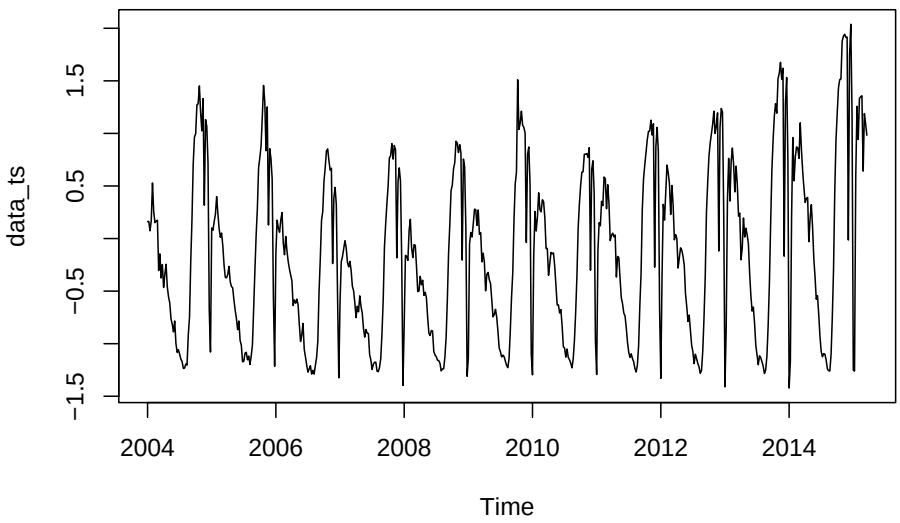
```r
library(ggplot2)
library(tseries)
library(forecast)
library(dplyr)
library(lubridate)
```

```r
# load data
data <- read.csv("/Users/ason/Desktop/courses/time series/Assignment/4/datasets/q1_trai
# transfer data to  time series format
data_ts <- ts(data$activity, start=c(2004,1), frequency=52)
```

```r
plot(data_ts)
```



```r
# Stationary test
ADF <- adf.test(data_ts)
```

```
## Warning in adf.test(data_ts): p-value smaller than printed p-value
```

```r
ADF
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  data_ts
## Dickey-Fuller = -8.0493, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

Since p-value is less than 0.05, we reject null hypothesis, i.e, this sequence is stationary.

```r
# White noise test
Box.test(data_ts, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  data_ts
## X-squared = 444.03, df = 1, p-value < 2.2e-16
```

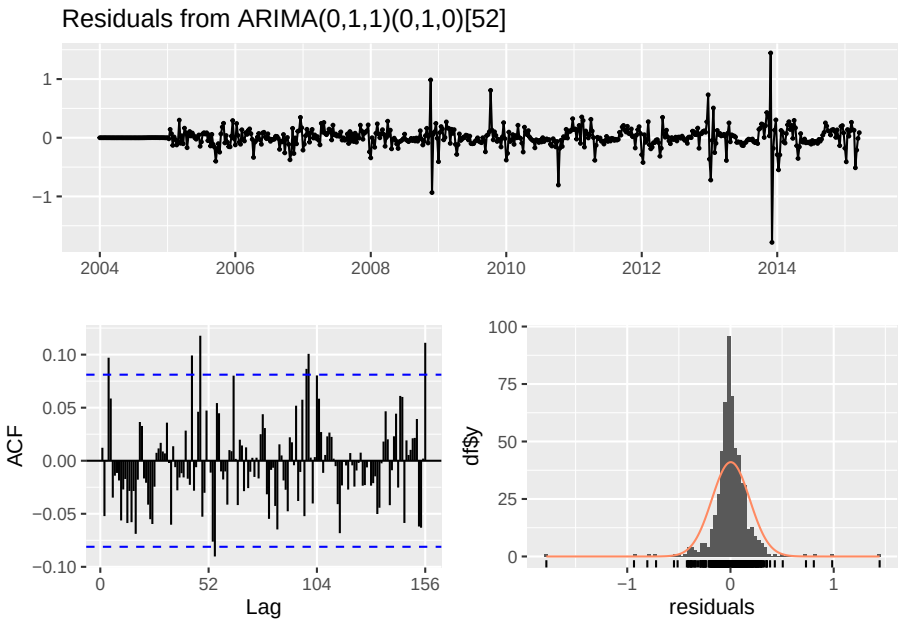
Since p-value is less than 0.05, we reject null hypothesis, i.e., this sequence is non-white noise and has analytical value.

```r
fit<-auto.arima(data_ts)
fit
```

```
## Series: data_ts
## ARIMA(0,1,1)(0,1,0)[52]
```

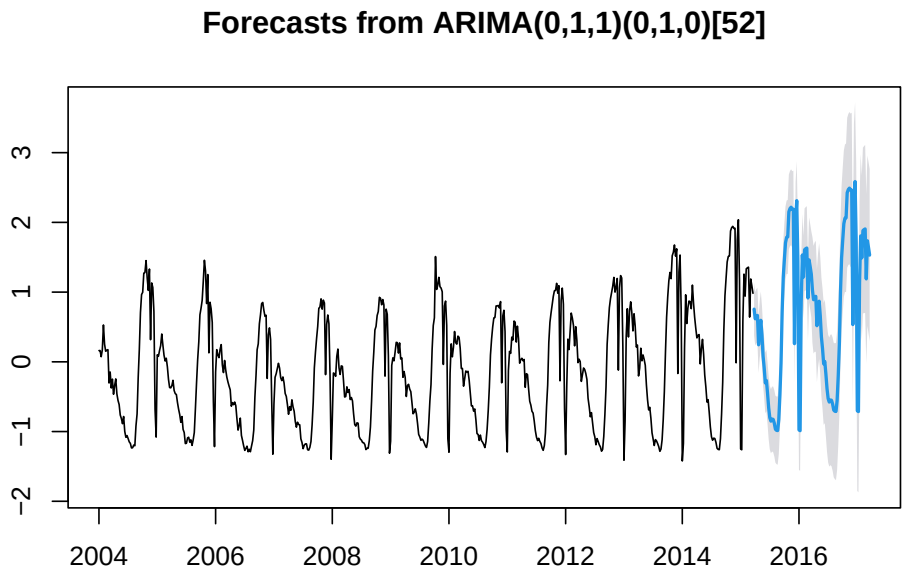
```
##
## Coefficients:
##           ma1
##       -0.8191
## s.e.   0.0284
##
## sigma^2 = 0.03699:  log likelihood = 121.9
## AIC=-239.8   AICc=-239.78   BIC=-231.25
```

```
fit1 <- Arima(data_ts,order=c(0,1,1),seasonal=c(0,1,0))
checkresiduals(fit1)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1)(0,1,0)[52]
## Q* = 113.62, df = 103, p-value = 0.223
##
## Model df: 1.   Total lags used: 104
```

```
futurVal <-forecast(fit,h = 104,level = c(95))
plot(futurVal)
```




```
# Extract the lower and upper bounds of the prediction intervals
lower <- futurVal$lower
upper <-futurVal$upper

# Extract the point forecasts
point_forecasts <- futurVal$mean

# Combine the lower bounds, point forecasts, and upper bounds into a data frame
output <- data.frame(lower, point_forecasts, upper)

# Write the data frame to a text file
#write.table(output, file = "predictions_q1.txt", sep = ",", row.names = FALSE, col.nam
```